Assessing spatial variability of nutrients, phytoplankton, and related water quality constituents in the California Sacramento-San Joaquin Delta at the landscape scale: 2018 High resolution mapping surveys

Methods

The study is based on the assessment and comparison of three broad-area high-resolution water quality mapping surveys of the Delta conducted in 2018, during the months of May, July and October. Each survey comprises three successive days of high-resolution mapping, with each day covering a different spatial extent; (i) one day covering the north-south axis of the Delta from the Sacramento River north of Freeport to the Harvey Banks export facility located in the southern Delta, including Old and Middle Rivers; (ii) one day covering the east-west axis of the Delta from the Sacramento River at Rio Vista to the San Joaquin River at Stockton, including Disappointment Slough and the North and South Forks of the Mokelumne River; and (iii) one day covering the northern and western Delta from the northern reaches of the Cache Slough Complex to the Grizzly, Honker, and Suisun Bays in the San Francisco Estuary. Data were collected at high speed to minimize to the extent to which tidal currents obscure nutrient and phytoplankton distributions.

Definition of regions and reaches for analysis

Aside from evaluation of the Delta as a whole, it is useful to divide the survey data into smaller sub-sets for analysis. We chose two main ways to do this. First, to evaluate temporal and spatial trends and to identify controls on nutrients, phytoplankton, and other water quality parameters at the regional scale, we defined eight *Study Regions* based on hydrodynamically distinct zones identified by Jon Burau (personal communication) and described in Brown et al. (in prep). Broadly speaking the Delta has three tidal transition zones (North Delta, Mokelumne System, South Delta), two strongly-tidally-forced tidal lagoons (Western and Central Delta), two dead-end channel systems (Cache Slough Complex, Suisun Marsh), and the brackish part of the estuary that connects the Delta to San Francisco Bay (Suisun Bay). The eight regions defined in this study and used in the online visualization tool thus include: (1) North Delta Tidal Transition Zone, (2) Mokelumne Tidal Transition Zone, (3) Cache Slough Complex, (4) Central Tidal Zone, (5) South Delta Tidal Transition Zone, (6) San Joaquin Tidal Transition Zone, (7) Western Tidal Zone, and (8) Suisun Bay. A ‘Transition Zone’ is where transport transitions from being mostly unidirectional, like in rivers, to mostly tidal. These eight regions are similar to regions proposed by Jabusch et al. (2016) to evaluate data collection efforts and nutrient status and trends analyses in the Delta; those subregions were based on Operational Landscape Units (OLUs) originally developed to represent restoration opportunity areas based on an understanding of ecological function, physical drivers, existing constraints, and elevation gradients, while taking into account hydrologic features, watershed boundaries, and DSM2 modeling requirements (Grenier and Grossinger, 2016).

For the purposes of this report and the associated data visualization, we also defined eight specific *Study Reaches* within these hydrodynamic regions. These study reaches were chosen to represent dominant flow paths that can be used to examine changes in nutrient concentrations and forms along with changes in phytoplankton abundances and species composition and represent a range of transport timescales from a few days to several weeks. These reaches are defined as (1) the Sacramento River, (2) the Lower Sacramento River, (3) Shag Slough, (4) Upper San Joaquin River, (5) Old River, (6) Middle River, (7) Hog Slough, and (8) Sycamore Slough. Data for these reaches are plotted against river mile, with the northern- or western-most point designated mile zero for each reach.

Sampling and analysis

Underway water sampling and onboard analyses:

Surveys were conducted using the USGS R/V Mary Landsteiner three times in 2018, each on three successive days: May 15, 16, and 17; July 24, 25, and 26; and October 17, 18, and 19. On each day data were collected during daylight hours beginning at approximately 7:30 am. For analysis, sample water was continuously pumped onto the boat while underway at speeds up to 30 mph (13 m/s) using a pick- up tube mounted at a fixed depth of approximately 1 m below the surface, routed through a screen to remove large debris and into a pressure-compensated manifold that maintained system pressure at a proscribed level irrespective of boat speed (Downing et al., 2016). A 2-stage debubbler was used to remove bubbles that can interfere with optical measurements. Sample water was split into separate flow paths for each analytical flow path, with constant flow rates maintained by metering the discharge line, except for the flow path serving the open split interface, which was metered on the feed line. Flows were continuously monitored using a sight gauge for each flow path.

The manifold delivered water to the following flowpaths: (1) A flow-through system consisting of a thermosalinograph that recorded temperature and conductance (Sea-Bird Scientific SB45 (TSG), Belluve, WA); fluorometers that measured fluorescence of chlorophyll-a (fCHLA) and fluorescence of dissolved organic matter (fDOM, WETLabs WETstar (WS), Philomath, OR); a beam transmissometer that recorded transmittance and attenuation (WETLabs model C-Star transmissometer (CStar), Philomath, OR), a nitrate (NO_3) analyzer (SUNA V2; Sea-Bird Scientific, Bellevue, WA). (2) A flow chamber for a multiparameter water quality sonde (YSI EXO2; Xylem Inc. (EXO), Rye Brook, New York) equipped with sensors to measure temperature, specific conductance (SpC), turbidity, pH, dissolved oxygen (DO), fDOM and fCHLA that then passed water to a fluorometer designed to measure different algal classes (FluoroProbe III (FP); BBE Moldaenke, Kiel, Germany). (3) An open-split interface at atmospheric pressure that served water filtered through a 0.45 micron high-capacity in-line groundwater sampling capsule filter (PALL Corporation, Port Washington, New York) to the on-board ammonium (NH_4) analyzer.

The NH_4 analyzer was a continuous flow, gas diffusion/conductivity-based (Carlson 1986) analyzer for ammonium (NH_4) analysis (TL-2800; Timberline Instruments, Boulder, CO) that was modified for field operation and continuous data collection by the manufacturer. Modifications included installation in a ruggedized housing, addition of an automated line-switching valve, addition of a heating unit to maintain the instrument at a constant above-ambient temperature, and changes to the software. Full standard curves were run at the beginning and end of each day and partial curves run throughout the day each time the boat stopped to sample. While underway, the analyzer was run in continuous mode with frequent periodic introduction of standard and blank solutions. The NH_4 analyzer was connected to a stand-alone computer and collected data through its native software.

All instrumentation was cleaned, and calibrations were checked prior to each use following the manufacturer's recommendation or as described below. Data for most instruments was recorded at 1- second frequency on a single data logger (CR6, Campbell Scientific, Logan, Utah, United States) together with a timestamp and boat position obtained from a high-resolution global positioning system receiver (16X-HVS, Garmin, Olathe, Kansas). The FluoroProbe logged data internally and to the host software. All data were displayed in real time, so scientists on board could respond when they observed changes relevant to study objectives, or if issues with flow or instrument performance were noted.

Discrete water sampling and laboratory analyses:

Discrete water samples were collected either underway or–in the case of more extensive sample requirements–while stopped on station. Discrete samples were collected while underway either using a fourth flow path on the sampling manifold or using a separate system with a separate pick-up tube mounted adjacent to the main sampling pick-up and pumping sample through a 0.45-micron filter into a sample bottle. Samples were collected while underway approximately every 2 miles, principally for determination of ortho-phosphate (PO_4). At designated sampling stations (approximately 10 per day) samples for determination of NH_4, NO_3, PO_4, dissolved organic carbon (DOC), and total dissolved nitrogen (TDN) concentrations were collected using this system. Additionally, unfiltered samples were collected approximately from 1 m depth using a submersible pump for analysis of chlorophyll-a (Chl-a), pheophytin, phycocyanin, allophycocyanin, and phycoerythrin pigments, phytoenumeration, and picocyanobacterial counts. Samples for phytoplankton enumeration and picocyanobacteria counts were fixed immediately with Lugol's iodine solution (4 mL) and formaldehyde (1 mL), respectively.

All samples were stored on ice in the dark during transit to the laboratory where they were stored at 2°C. Dissolved organic carbon samples were preserved to <2 pH with high-purity sulphuric acid the day following collection and stored at 2°C. Samples for pigment analysis were filtered ours of collection and immediately frozen (-90°C). Total Chl-a samples were filtered through a 0.7 µm nominal pore size glass fiber filter (Advantec MFS, Inc, Dublin, CA). Chlorophyll-a associated with the larger sized cell fraction were filtered through a 5 µm nominal pore size 25 mm polycarbonate filter (General Electric Healthcare, Chicago, IL). Samples for phycocyanin, allophycocyanin, and phycoerythrin pigment analysis were filtered through a 0.3 µm nominal pore size glass fiber filter (Advantec MFS, Inc, Dublin, CA). Frozen Chl- a samples and chilled nutrient samples were shipped to the USGS National Water Quality Laboratory (Denver, CO). Frozen filters for analysis of phycocyanin, allophycocyanin, and phycoerythrin were shipped to BSA Environmental Services, Inc. (Beachwood, OH) as were preserved picocyanobacteria and phytoplankton enumeration samples.

Concentrations of nutrients were determined colorimetrically using an automated-segmented flow analyzer. Concentrations of nitrogen as nitrite (NO_2-N) and as nitrate plus nitrite (NO_3-N + NO_2-N) were determined by colorimetric analysis (Fishman, 1993; Patton and Kryskalla, 2011), ammonium as nitrogen (NH_4-N) was determined by colorimetric analysis after reaction with salicylate-hypochlorite (Fishman, 1993), and total dissolved nitrogen (TDN) was determined by alkaline persulfate digestion (Patton and Kryskalla 2003). Orthophosphate as phosphorous (PO_4-P; also referred to as soluble reactive phosphate) was determined by colorimetric analysis after reaction with NH_4 molybdate and reduction with ascorbic acid (Patton and Truitt, 1992). Chlorophyll-a and phaeophytin concentrations were determined according to EPA Method 445.0 (Arar and Collins, 1997) using sonication extraction.

Dissolved organic carbon concentrations were measured using a total organic carbon analyzer (TOC-V_{CSH}, Shimadzu Scientific Instruments, Columbia, Maryland) using high-temperature catalytic combustion according to a modified version of USEPA method 415.3 (Potter and Wimsatt, 2005). Absorbance spectra and fluorescence EEMs were simultaneously measured using an Aqualog® equipped with a photodiode and charge-coupled device (CCD) (Horiba Instruments, NJ, USA) according to the method described by Hansen et al. (2018). Briefly, excitation and absorbance spectra were measured using a 150 W xenon lamp, a 5 nm bandpass, and a 1 s integration time at wavelengths of 240–600 nm. Emission spectra were collected with a CCD at approximately 1.64 nm (4 pixel) intervals at wavelengths of 250–600 nm. To reduce ultraviolet light exposure of the sample and limit the effects of photobleaching, excitation and absorbance wavelengths were scanned from low to high energy (i.e., visible to ultraviolet range) during analysis. Peaks, ratios, and slopes reported in the data were calculated as described by Hansen et al. (2016).

Phycocyanin, allophycocyanin, and phycoerythrin were determined by azolectin-CHAPS buffer extraction as described by Zimba (2012) and modified from Bennett and Bogorad (1973) at BSA Environmental Services. Phytoplankton enumeration was completed in accordance with the American Public Health Association (APHA) Standard Method 10200 (2012). Picocyanobacteria direct counts were enumerated via epifluorescence microscopy as described by Murrell and Lores (2004). Phycoerythrin-containing cells (PE) and phycocyanin-containing cells (PC) were counted separately and then summed to obtain total picocyanobacteria densities for cells sized between 0.2 and 2.0 microns.

Data processing

Data from onboard continuous instruments not directly logged to the flow-through data collection system were merged based on timestamp to the nearest second. All data directly logged to the flow-through data collection system was processed using pandas software library in Python to remove periods of compromised data (e.g. flow blockages, bubbles), to apply instrument corrections and unit conversions, and to apply a centered 20 second median filter to the time series. Onboard NO_3 data was processed to remove the salinity-dependent interference of bromide and DOM using the Sakamoto et al. (2009) method (as implemented in the Seabird UCI software). Final NO_3 values were obtained by regressing instrument response against NO_3 concentrations obtained from laboratory measurements of discrete samples collected through the course of the day. Individual sample results with more than three standard deviations from the regression were judged to be outliers and removed from the regression.

No more than one outlier was removed from any day. NH_4 data was processed by first correcting the voltage output of the instrument for baseline drift based on the instrument response during periodic measurements of organic-free deionized water (n ≥10) and then using a regression model of the voltage response to a standard concentration. A regression coefficient of >0.97 R^2 or better was judged to be acceptable. The NH4 data timestamp was corrected to account for the time lag associated with the microfluidic processing prior to integration with other data sources.

Dissolved inorganic nitrogen (DIN) concentration was calculated as the sum of NO_3, NO_2 and NH_4– both for the continuous data and the discrete sample data. Laboratory DON concentration was calculated by subtracting NH_4, NO_3, and NO_2 data from the TDN data (i.e. DON = TDN – DIN). Continuous DOC and DON concentrations were developed by regressing onboard fDOM data against the DOC and DON concentrations measured in discrete samples. Data exceeding 3 times the standard deviation of the residuals was judged to be an outlier and not included in the model. Total dissolved nitrogen (TDN) was calculated as the sum of DIN (NO_3 + NH_4) and DON.

Data was spatially aligned (SA) to a common spatial framework to facilitate comparisons between dates and to minimize the effects of differential data density on the spatial interpolation calculations by assigning median-filtered data by proximity to a centerline vector comprising points located at the centerline of all channels navigated, spaced at approximately 150 m intervals. Interpolated water quality maps were created from the spatially aligned data using ArcGIS Spline with Barriers tool (Terzopoulos and Witkin, 1988). The raster output was smoothed with the Focal Statistics tool, and interpolated values were extracted to the spatially aligned points to create continuous spatially aligned data across the mapping domain.

References

Arar, E.J. and Collins, G.B. (1997) Method 445.0: In vitro determination of chlorophyll a and pheophytin a in marine and freshwater algae by fluorescence. United States Environmental Protection Agency, Office of Research and Development, National Exposure Research Laboratory.

Bennett, A. and Bogorad, L. (1973) Complementary chromatic adaptation in a filamentous blue-green alga. The Journal of cell biology 58, 419-435.

Carlson, R.M. (1986). Continuous flow of reduction of nitrate to ammonia with granular zinc. Analytical Chemistry 58, 7, 1590-1591.

Downing, B.D., Bergamaschi, B.A., Kendall, C., Kraus, T.E.C., Dennis, K.J., Carter, J.A. and Von Dessonneck, T.S. (2016) Using Continuous Underway Isotope Measurements to Map Water Residence Time in Hydrodynamically Complex Tidal Environments. Environmental Science & Technology 50, 13387-13396.

Fishman, M.J. (1993) Methods of analysis by the US Geological Survey National Water Quality Laboratory: Determination of inorganic and organic constituents in water and fluvial sediments. US Geological Survey open-file report (USA).

Grenier, L., Grossinger, R. (2016) Tools for landscape-scale restoration planning in the Delta. Presentation to the Delta Independent Science Board, January 16, 2013, 2nd Floor Conference Center, Sacramento, CA.

Hansen, A.M., Kraus, T.E.C., Pellerin, B. A., Fleck, J. A., Downing, B. D., Bergamaschi, B. A. (2016) Optical properties of dissolved organic Matter (DOM): effects of biological and photolytic degradation:

Limnology and Oceanography, v. 61, p. 1015-1032, https://doi.org/10.1002/lno.10270.

Hansen, A. M., Fleck, J. A., Kraus, T. E. C., Downing, B. D., Von Dessonneck, T., Bergamaschi, B. A. (2018) Procedures for using the Horiba Scientific Aqualog® fluorometer to measure absorbance and fluorescence from dissolved organic matter: U. S. Geological Survey Open-File Report 2018-1096, 31 p., https://doi.org/10.3133/ofr20181096.

Jabusch, T., Bresnahan, P., Trowbridge, P., Novick, E., Wong, A., Salomon, M. and Senn, D. (2016) Summary and evaluation of Delta subregions for nutrient monitoring and assessment Technical report. San Francisco Estuary Institute, Richmond, CA. [Google Scholar].

Murrell, M.C. and Lores, E.M. (2004) Phytoplankton and zooplankton seasonal dynamics in a subtropical estuary: importance of cyanobacteria. Journal of Plankton Research 26, 371-382.

Patton, C.J. and Kryskalla, J.R. (2011) Colorimetric determination of nitrate plus nitrite in water by enzymatic reduction, automated discrete analyzer methods. US Geological Survey Techniques and Methods 34.

Patton, C.J. and Truitt, E.P. (1992) Methods of Analysis by the US Geological Survey National Water Quality Laboratory: Determination of the Total Phosphorus by a Kjeldahl Digestion Method and an Automated Colorimetric Finish that Includes Dialysis. US Geological Survey.

Potter, B. and Wimsatt, J. (2005) Determination of total organic carbon and specific UV absorbance at 254 nm in source water and drinking water. EPA Document: EPA.

Sakamoto, C.M., Johnson, K.S. and Coletti, L.J. (2009) Improved algorithm for the computation of nitrate concentrations in seawater using an in situ ultraviolet spectrophotometer. Limnology and Oceanography: Methods 7, 132-143.

Terzopoulos, D. and Witkin, A. (1988) Physically based models with rigid and deformable components. IEEE Computer Graphics and Applications 8, 41-51.

Zimba, P.V. (2012) An improved phycobilin extraction method. Harmful Algae 17, 35-39.